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CS1 C12

ESSENTIALS OF DATA SCIENCE

PRACTICE LAB ASSIGNMENT

4.1.1. PANDAS - SERIES CREATION AND MANIPULATION

WriteaPython program that takesa listofnumbersfrom theuser, createsa Pandasseriesfromit,andthen calculates the mean of even andodd numbersseparately using the groupby andmean()operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

INPUT

```python
import pandas as pd

# Take inputs from the user to create a list of numbers
numbers = list(map(int, input().split()))

# Create a Pandas series from the list of numbers
series=pd.Series(numbers)
# Grouping by even and odd numbers and calculating the mean
grouped =series.groupby(series%2==0).mean()

# Display the mean of even and odd numbers with labels
grouped.index = ['Even' if is_even else 'Odd' for is_even in
grouped.index]
print("Mean of even and odd numbers:")
print(grouped)
```

OUTPUT

Average time	Maximum time
0.029 s	0.031 s
29.00 ms	31.00 ms

3 out of 3 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 31 ms Debug

Expected output	Actual output
1 2 3 4 5 6 7 8 9 10	1 2 3 4 5 6 7 8 9 10
Mean of even and odd numbers:	Mean of even and odd numbers:
Odd 5.0	Odd 5.0
Even 6.0	Even 6.0
dtype: float64	dtype: float64

Test case 2 28 ms Debug

Expected output	Actual output
4 4 4 6 6 2 2 2 10 12 14 16	4 4 4 6 6 2 2 2 10 12 14 16
Mean of even and odd numbers:	Mean of even and odd numbers:
Even 6.833333	Even 6.833333
dtype: float64	dtype: float64

Test case 3 28 ms Debug

Expected output	Actual output
1 5 7 9 11 13 15 17 111 21 25	1 5 7 9 11 13 15 17 111 21 25
Mean of even and odd numbers:	Mean of even and odd numbers:
Odd 21.363636	Odd 21.363636
dtype: float64	dtype: float64

4.1.2. DICTIONARY TO DATAFRAME

Adictionary oflists has been providedtoyouintheeditor.CreateaDataFramefromthedictionaryoflistsand perform the listed operations, then display the DataFrame before and after eachmanipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column Gender with values taken from the user.
- Display the DataFrame after adding the new column.

Modify a column:

- Convert names to uppercase.
- Display the DataFrame after modifying the column.

Delete a column:

- Remove the Age column.
- Display the DataFrame after deleting the column.

INPUT

```python
import pandas as pd

# Provided dictionary of lists
data = {
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Age': [25, 30, 35],
}

# Convert the dictionary to a DataFrame
df = pd.DataFrame(data)

# Display the original DataFrame
print("Original DataFrame:")
print(df)
name=input("New name: ")
age=int(input("New age: "))
# Adding a new row
new_data={"Name":name,"Age":age}

df=df.append(new_data,ignore_index=True)
#df=pd.concat([df,pd.DataFrame([new_data])],ignore_index=True
)


# Display the DataFrame after adding a new row
print("After adding a row:\n",df)

# Modifying a row
i=int(input("Index of row to modify: "))
new_age=int(input("New age: "))
df.loc[i,"Age"]=new_age



# Display the DataFrame after modifying a row
print("After modifying a row:")
print(df)

# Deleting a row
i=int(input("Index of row to delete: "))
df=df.drop(df.index[i]).reset_index(drop=True)
# Display the DataFrame after deleting a row
print("After deleting a row:")
print(df)

# Adding a new column
g_input=input("Enter genders separated by space: ")
genders=g_input.split()
df["Gender"]=genders

# Display the DataFrame after adding a new column
print("After adding a new column:")
print(df)

# Modifying a column

# Display the DataFrame after modifying a column
df["Name"]=df["Name"].str.upper()
print("After modifying a column:")
print(df)
```

OUTPUT

Average time	Maximum time	
0.115 s	0.119 s	1 out of 1 shown test case(s) passed
114.50 ms	119.00 ms	1 out of 1 hidden test case(s) passed

Test case 1 119 ms Debug

Expected output

```
Original DataFrame:
      Name  Age
0    Alice   25
1      Bob   30
2  Charlie   35
New name: Susan
New age: 29
After adding a row:
       Name  Age
0    Alice   25
1      Bob   30
2  Charlie   35
3    Susan   29
Index of row to modify: 0
New age: 27
After modifying a row:
      Name  Age
0    Alice   27
1      Bob   30
2  Charlie   35
3    Susan   29
Index of row to delete: 3
After deleting a row:
      Name  Age
0    Alice   27
1      Bob   30
2  Charlie   35
Enter genders separated by space: F M M
After adding a new column:
      Name  Age Gender
0    Alice   27      F
1      Bob   30      M
2  Charlie   35      M
After modifying a column:
      Name  Age Gender
0    ALICE   27      F
1      BOB   30      M
2  CHARLIE   35      M
After deleting a column:
      Name Gender
0    ALICE      F
1      BOB      M
2  CHARLIE      M
```

Actual output

```
Original DataFrame:
      Name  Age
0    Alice   25
1      Bob   30
2  Charlie   35
New name: Susan
New age: 29
After adding a row:
       Name  Age
0    Alice   25
1      Bob   30
2  Charlie   35
3    Susan   29
Index of row to modify: 0
New age: 27
After modifying a row:
      Name  Age
0    Alice   27
1      Bob   30
2  Charlie   35
3    Susan   29
Index of row to delete: 3
After deleting a row:
      Name  Age
0    Alice   27
1      Bob   30
2  Charlie   35
Enter genders separated by space: 
M
After adding a new column:
      Name  Age Gender
0    Alice   27      F
1      Bob   30      M
2  Charlie   35      M
After modifying a column:
      Name  Age Gender
0    ALICE   27      F
1      BOB   30      M
2  CHARLIE   35      M
After deleting a column:
      Name Gender
0    ALICE      F
1      BOB      M
2  CHARLIE      M
```

4.1.3. STUDENT INFORMATION

Writeaprogram to read atext filecontainingstudentinformation(name,age,andgrade)usingPandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

INPUT

```
import pandas as pd

# Read the text file into a DataFrame
file = input()
data = pd.read_csv(file, sep="\s+", header=None, names=
["Name", "Age", "Grade"])


# write your code here..
df=pd.DataFrame(data)
print("First five rows:")
print(df.head(5))
print("Average age:",df["Age"].mean().round(2))
print("Students with a grade up to B")
filtered_df=df[df['Grade'].isin(['A', 'B'])]
print(filtered_df)
```

OUTPUT

Average time	Maximum time	
0.034 s 34.00 ms	0.040 s 40.00 ms	1 out of 1 shown test case(s) passed 1 out of 1 hidden test case(s) passed

Test case 1 40 ms Debug

Expected output	Actual output
studentdata.txt	studentdata.txt
First five rows:	First five rows:
Name Age Grade	Name Age Grade
0 John 25 A	0 John 25 A
1 Alin 22 B	1 Alin 22 B
2 Emma 24 A	2 Emma 24 A
3 Mike 23 C	3 Mike 23 C
4 Sony 21 B	4 Sony 21 B
Average age: 23.29	Average age: 23.29
Students with a grade up to B	Students with a grade up to B
Name Age Grade	Name Age Grade
0 John 25 A	0 John 25 A
1 Alin 22 B	1 Alin 22 B
2 Emma 24 A	2 Emma 24 A
4 Sony 21 B	4 Sony 21 B
5 Amia 26 A	5 Amia 26 A

LAB ASSIGNMENT

4.2.1. MONTH WITH THE HIGHEST TOTAL SALES

Writea Python program that takes the file name ofa CSV file as input, readsthedata,andperformsthefollowingoperations:

- The CSV file contains the columns: Date, Product, Quantity, Price, andCity.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

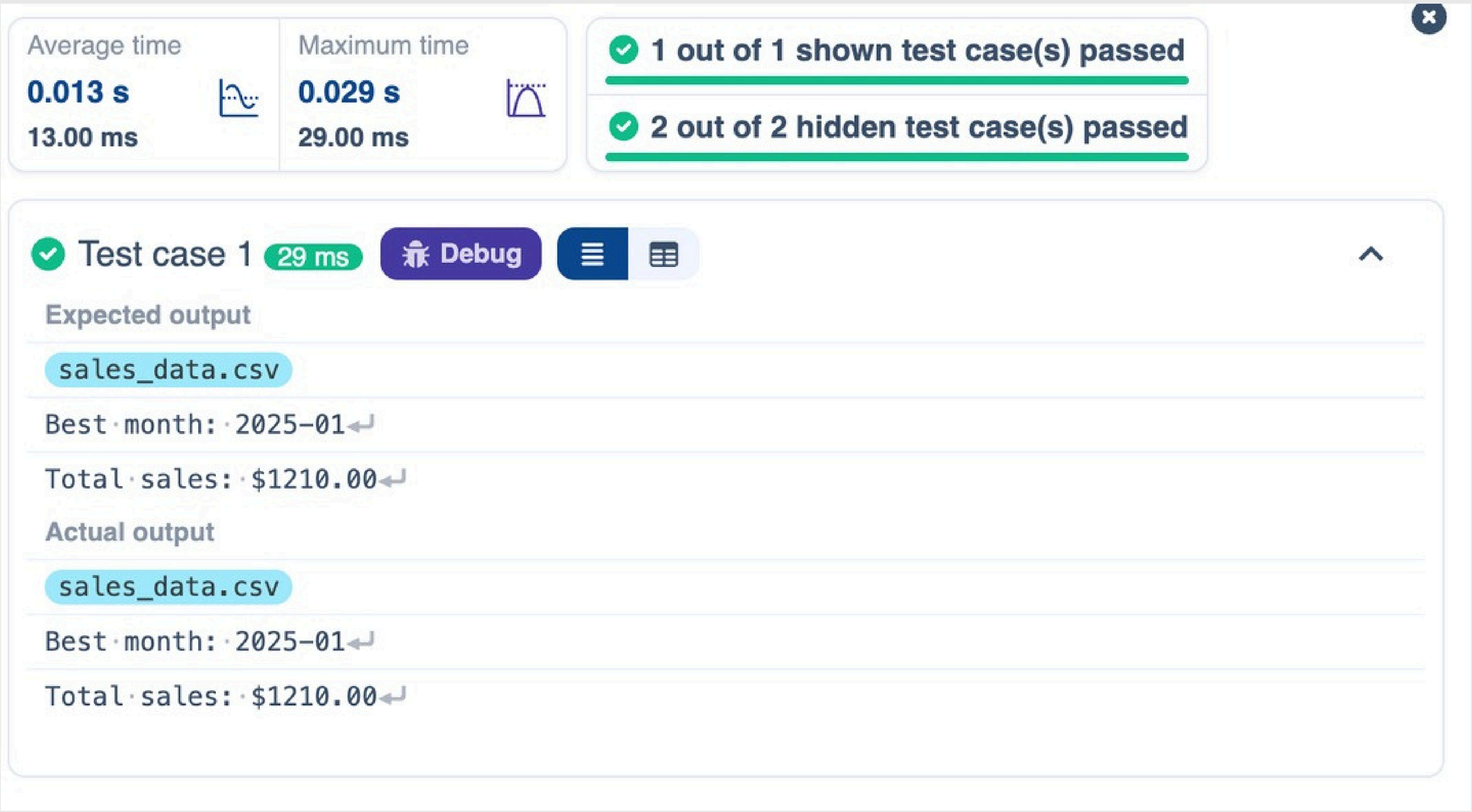
Sample Data:

Date,Product ,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles

INPUT

```python
import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)



df['Date'] = pd.to_datetime(df['Date'])

# Calculate total sales for each transaction
df['Total_Sales'] = df['Quantity'] * df['Price']

# Extract month and year from Date
df['Month'] = df['Date'].dt.to_period('M')  # Creates YYYY-MM format

# Find the month with the highest total sales
monthly_sales = df.groupby('Month')['Total_Sales'].sum()
best_month = monthly_sales.idxmax()
highest_sales =monthly_sales.max()


print(f"Best month: {best_month}")
print(f"Total sales: ${highest_sales:.2f}")
```

OUTPUT



4.2.2. BEST SELLING PRODUCT

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

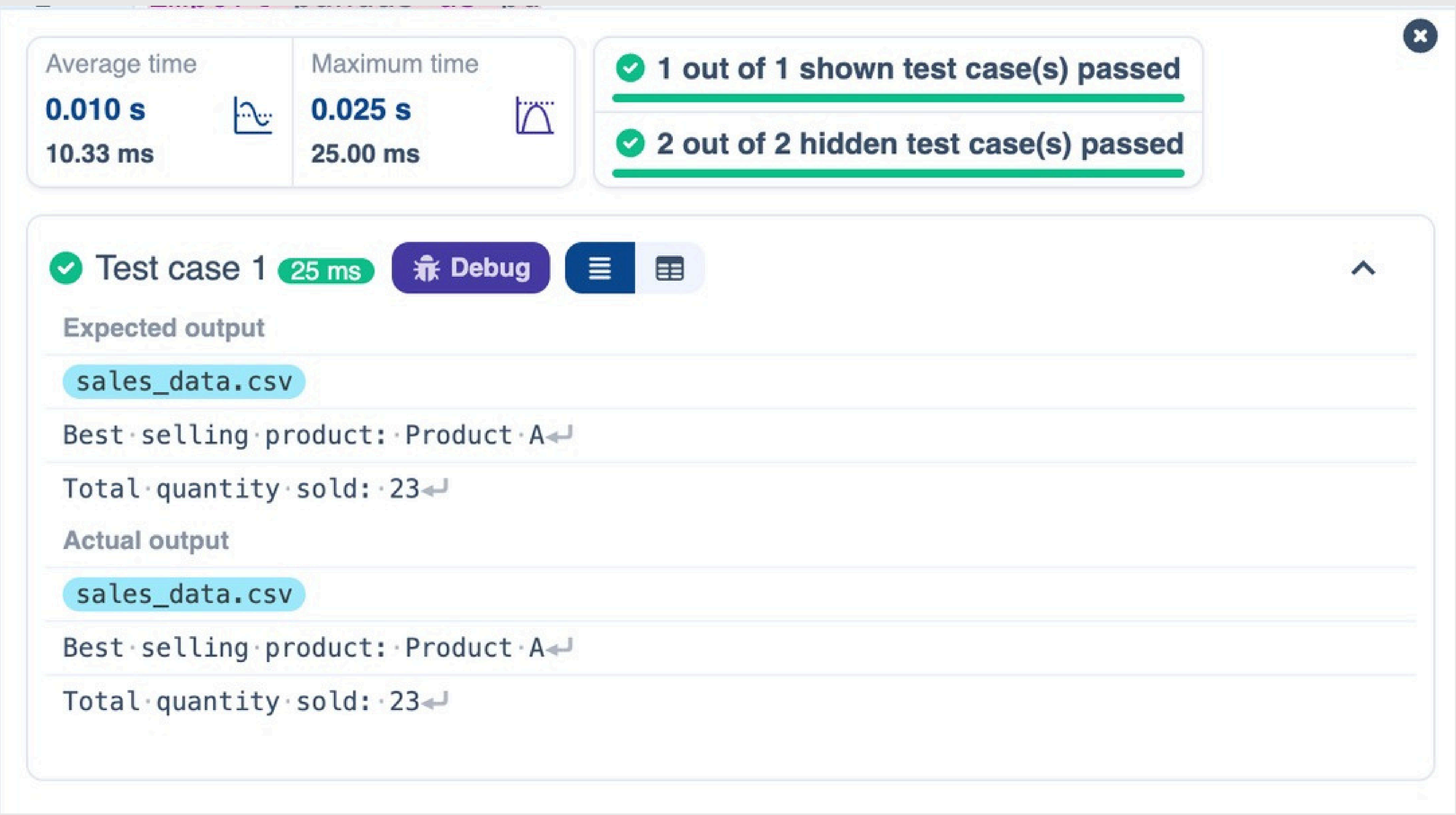
Sample Data:

```
Date,Product ,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

INPUT

```python
import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)


product_quantity = df.groupby('Product')['Quantity'].sum()

best_product = product_quantity.idxmax()
highest_quantity = product_quantity.max()

# Display the result
print(f"Best selling product: {best_product}")
print(f"Total quantity sold: {highest_quantity}")
```

OUTPUT



4.2.3. CITY THAT SOLD THE MOST PRODUCTS

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

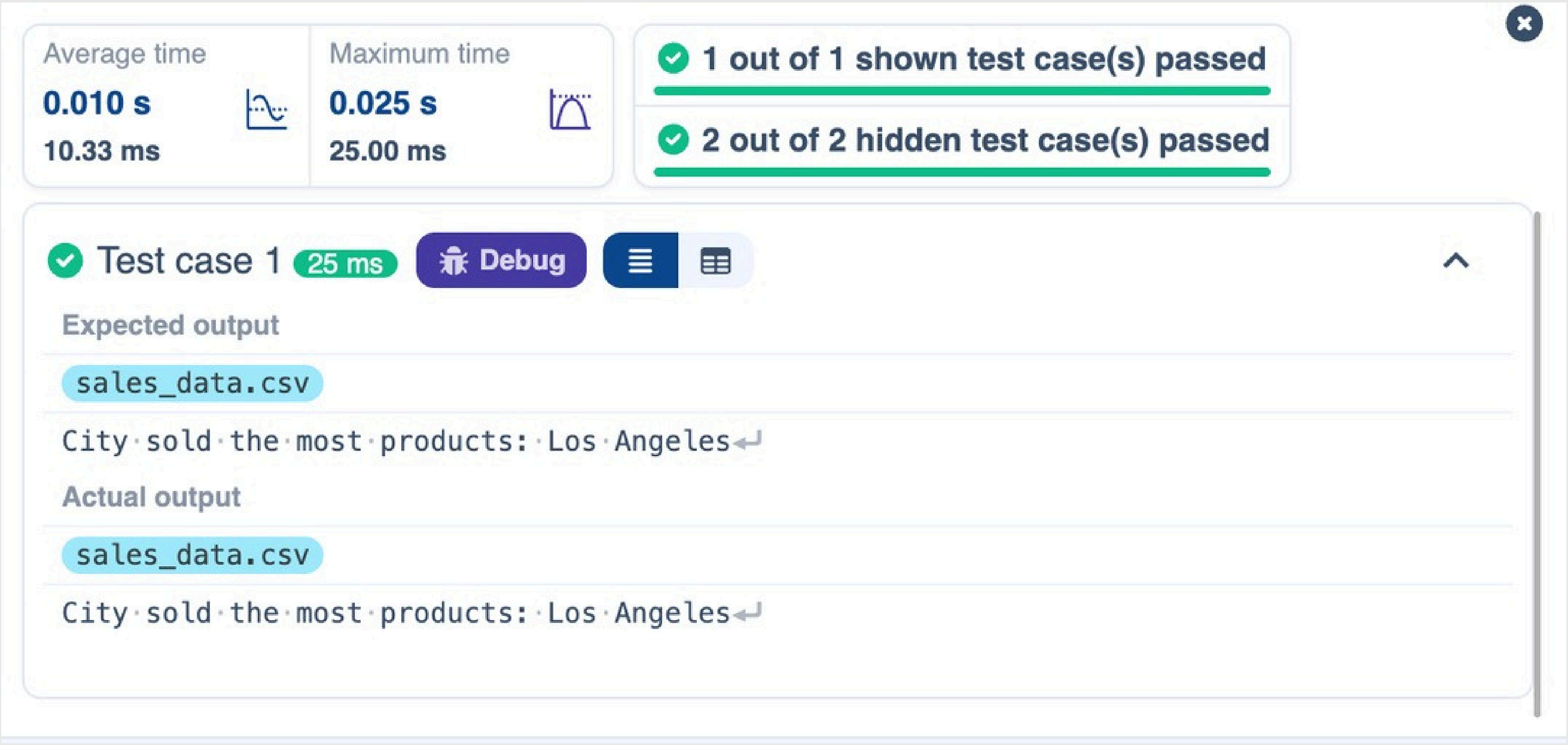
Sample Data:

```
Date,Product ,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

INPUT

```python
import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)

city_sales = df.groupby('City')['Quantity'].sum()

best_city = city_sales.idxmax()

# Display the result
print(f"City sold the most products: {best_city}")
```

OUTPUT



4.2.4. MOST FREQUENTLY SOLD PRODUCT PAIRS

Writea Python program that takes the file name ofa CSV fileas input,reads the data, and performsthe followingoperations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, andCity.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pair/s that was sold most frequently.

Sample Data:

Date,Product ,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
Explanation:
Transactions:

- 2025-01-01: Product A, Product B
- 2025-01-02: Product A, Product C
- 2025-01-03: Product B, Product A
- 2025-01-04: Product C, Product B
- 2025-01-05: Product A, Product C

Now, let's count how often the pairs of products appear together:

- Product A and Product B: Appear in transactions on 2025-01-01 and 2025-01-03.
- Product A and Product C: Appear in transactions on 2025-01-02 and 2025-01-05.
- Product B and Product C: Appears in transactions on 2025-01-04.

Most Frequent Product Combinations:

- Product A and Product B (2 times)
- Product A and Product C (2 times)

INPUT

```python
import pandas as pd
from itertools import combinations
from collections import Counter

# Prompt user to input the file name
file_name = input()

# Read data from the specified CSV file
df = pd.read_csv(file_name)


daily_transactions = df.groupby('Date')['Product'].apply(list)
# Initialize a Counter to store the frequency of each pair
pair_counts = Counter()
# 2. Iterate through each date's transaction
for products in daily_transactions:
# Sort the products to ensure (A, B) is treated the same as (B, A)
    products = sorted(products)

    pairs = list(combinations(products, 2))
# Update the counter with these pairs
    pair_counts.update(pairs)

if pair_counts:

    max_count = max(pair_counts.values())
    for pair, count in pair_counts.items():
        if count == max_count:
            print(f"{pair[0]} and {pair[1]}: {count} times")
else:
    print("No product pairs found.")
```

OUTPUT

Average time
0.010 s
9.67 ms

Maximum time
0.022 s
22.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 22 ms Debug

Expected output

sales_data.csv

Product A and Product B: 2 times

Product A and Product C: 2 times

Actual output

sales_data.csv

Product A and Product B: 2 times

Product A and Product C: 2 times

4.2.5. TITANIC DATASET ANALYSIS AND DATA CLEANING

YouareprovidedwiththeTitanicdatasetcontaininginformationaboutpassengersontheTitanic.YourtaskistowritePythoncodetoanswerthefollowingquestions basedonthedataset. For eachquestion,performnecessarydatacleaning,transformations,andcalculationsasrequired.

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows of the dataset.
3. Get the shape of the dataset (number of rows and columns).
4. Get a summary of the dataset (using .info()).
5. Get basic statistics (mean, standard deviation, etc.) of the dataset using .describe().
6. Check for missing values and display the count of missing values for each column.
7. Fill missing values in the ‘Age’ column with the median age.
8. Fill missing values in the ‘Embarked’ column with the most frequent value (mode()).
9. Drop the ‘Cabin’ column due to many missing values.
10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

INPUT

```python
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# 1. Display the first 5 rows of the dataset
print(data.head())
# 2. Display the last 5 rows of the dataset
print(data.tail())
# 3. Get the shape of the dataset
print(data.shape)
# 4. Get a summary of the dataset (info)
print(data.info())
# 5. Get basic statistics of the dataset
print(data.describe())
# 6. Check for missing values
print(data.isnull().sum())
# 7. Fill missing values in the ‘Age’ column with the median age
data['Age'].fillna(data['Age'].median(), inplace = True)
# 8. Fill missing values in the ‘Embarked’ column with the mode
data['Embarked'].fillna(data['Embarked'].mode()
[0],inplace=True)
# 9. Drop the ‘Cabin’ column due to many missing values
data.drop(columns='Cabin',inplace = True)
# 10. Create a new column 'FamilySize' by adding 'SibSp' and 'Parch'
data["FamilySize"] =data['SibSp']+data['Parch']
```

OUTPUT

Average time 0.069 s 69.00 ms | Maximum time 0.069 s 69.00 ms | 1 out of 1 shown test case(s) passed

Test case 1 69 ms Debug

Expected output

```
   PassengerId  Survived  Pclass  ...     Fare Cabin  Embarked
0            1         0       3  ...   7.2500   NaN         S
1            2         1       1  ...  71.2833   C85         C
2            3         1       3  ...   7.9250   NaN         S
3            4         1       1  ...  53.1000  C123         S
4            5         0       3  ...   8.0500   NaN         S

[5 rows x 12 columns]
     PassengerId  Survived  Pclass  ...   Fare Cabin  Embarked
886          887         0       2  ...  13.00   NaN         S
887          888         1       1  ...  30.00   B42         S
888          889         0       3  ...  23.45   NaN         S
889          890         1       1  ...  30.00  C148         C
890          891         0       3  ...   7.75   NaN         Q

[5 rows x 12 columns]
(891, 12)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object 
 11  Embarked     889 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 66.2+ KB
None
       PassengerId    Survived      Pclass  ...       SibSp       Parch        Fare
count   891.000000  891.000000  891.000000  ...  891.000000  891.000000  891.000000
mean    446.000000    0.383838    2.308642  ...    0.523008    0.381594   32.204208
std     257.353842    0.486592    0.836071  ...    1.102743    0.806057   49.693429
min       1.000000    0.000000    1.000000  ...    0.000000    0.000000    0.000000
25%     223.500000    0.000000    2.000000  ...    0.000000    0.000000    7.910400
50%     446.000000    0.000000    3.000000  ...    0.000000    0.000000   14.454200
75%     668.500000    1.000000    3.000000  ...    1.000000    0.000000   31.000000
max     891.000000    1.000000    3.000000  ...    8.000000    6.000000  512.329200

[8 rows x 7 columns]
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

Actual output

```
   PassengerId  Survived  Pclass  ...     Fare Cabin  Embarked
0            1         0       3  ...   7.2500   NaN         S
1            2         1       1  ...  71.2833   C85         C
2            3         1       3  ...   7.9250   NaN         S
3            4         1       1  ...  53.1000  C123         S
4            5         0       3  ...   8.0500   NaN         S

[5 rows x 12 columns]
     PassengerId  Survived  Pclass  ...   Fare Cabin  Embarked
886          887         0       2  ...  13.00   NaN         S
887          888         1       1  ...  30.00   B42         S
888          889         0       3  ...  23.45   NaN         S
889          890         1       1  ...  30.00  C148         C
890          891         0       3  ...   7.75   NaN         Q

[5 rows x 12 columns]
(891, 12)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object 
 11  Embarked     889 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 66.2+ KB
None
       PassengerId    Survived      Pclass  ...       SibSp       Parch        Fare
count   891.000000  891.000000  891.000000  ...  891.000000  891.000000  891.000000
```

4.2.6. TITANIC DATASET ANALYSIS AND DATA CLEANING - 2

Youareprovided withthe Titanic dataset containinginformationaboutpassengers on theTitanic.YourtaskistowritePythoncodetoanswerthefollowingquestions basedon the dataset.

1. Create a new column ‘IsAlone’ which is 1 ifthepassengeris alone(FamilySize= 0), otherwise0.
2. Convert the ‘Sex' column to numeric values (male: 0, female: 1).
3. One-hot encode the ‘Embarked’ column, dropping the first category.
4. Get the mean age of passengers.
5. Get the median fare of passengers.
6. Get the number of passengers by class.
7. Get the number of passengers by gender.
8. Get the number of passengers by survival status.
9. Calculate the survival rate of passengers.
10. Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

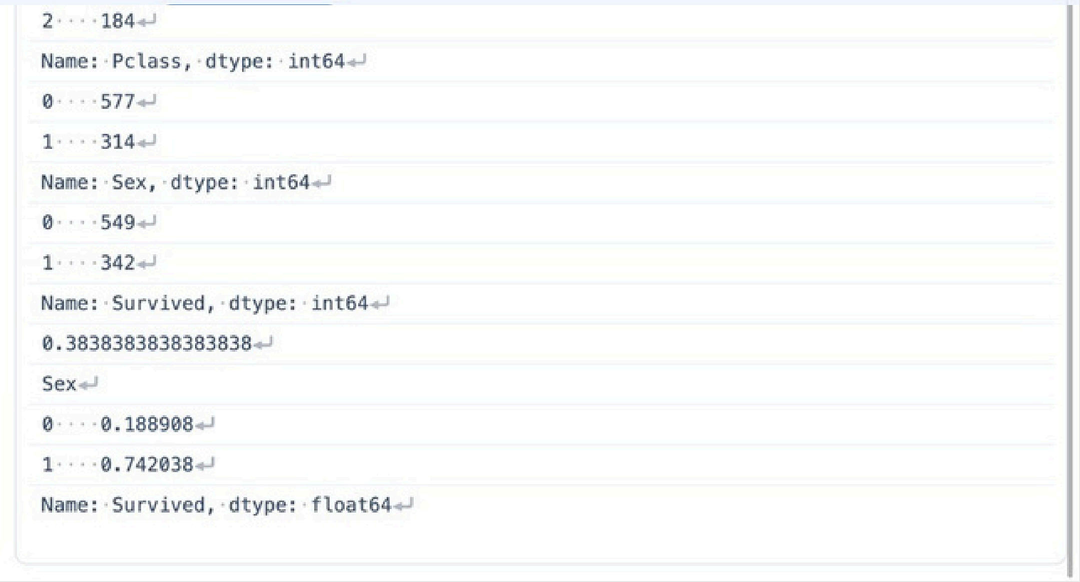
Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

INPUT

```
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data['FamilySize'] = data['SibSp'] + data['Parch']

# Create a new column 'IsAlone' which is 1 if the passenger
is alone

data['IsAlone'] = np.where(data['FamilySize']== 0, 1, 0)

# 2. Convert 'Sex' to numeric (male: 0, female: 1)
data['Sex'] = data['Sex'].map({'male': 0,'female': 1})
# 3. One-hot encode the 'Embarked' column
data = pd.get_dummies(data, columns=['Embarked'])
# 4. Get the mean age of passengers
mean_age = data['Age'].mean()
print( mean_age)
# 5. Get the median fare of passengers
median_fare = data['Fare'].median()
print( median_fare)
# 6. Get the number of passengers by class
print( data['Pclass'].value_counts())
# 7. Get the number of passengers by gender
print( data['Sex'].value_counts())
# 8. Get the number of passengers by survival status
print( data['Survived'].value_counts())
# 9. Calculate the overall survival rate
survival_rate = data['Survived'].mean()
print(format(survival_rate))
# 10. Calculate the survival rate by gender
survival_by_gender = data.groupby('Sex')['Survived'].mean()
print( survival_by_gender)
```

OUTPUT



4.2.7. TITANIC DATASET ANALYSIS AND DATA CLEANING - 3

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions basedonthe dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).
8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

Pas sen ger Id	Sur viv ed	Pcl ass	Na me	Sex	Ag e	Sib Sp	Par ch	Tic ket	Far e	Ca bin	Em bar ked

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

INPUT

```python
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data['FamilySize'] = data['SibSp'] + data['Parch']
data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
data = pd.get_dummies(data, columns=['Embarked'],
drop_first=True)

# 1. Calculate the survival rate by class
print(data.groupby('Pclass')['Survived'].mean())
# 2. Calculate the survival rate by embarkation location
(Embarked_S)
print(data.groupby('Embarked_S')['Survived'].mean())
# 3. Calculate the survival rate by family size
print(data.groupby('FamilySize')['Survived'].mean())
# 4. Calculate the survival rate by being alone
print(data.groupby('IsAlone')['Survived'].mean())
# 5. Get the average fare by passenger class
print(data.groupby('Pclass')['Fare'].mean())
# 6. Get the average age by passenger class
print(data.groupby('Pclass')['Age'].mean())
# 7. Get the average age by survival status
print(data.groupby('Survived')['Age'].mean())
# 8. Get the average fare by survival status
print(data.groupby('Survived')['Fare'].mean())
# 9. Get the number of survivors by class
# Filter for Survived=1, then count occurrences of Pclass
print(data[data['Survived'] == 1]['Pclass'].value_counts())
# 10. Get the number of non-survivors by class
# Filter for Survived=0, then count occurrences of Pclass
print(data[data['Survived'] == 0]['Pclass'].value_counts())
```

OUTPUT

Average time 0.038 s 38.00 ms | Maximum time 0.038 s 38.00 ms | 1 out of 1 shown test case(s) passed

Test case 1 38 ms Debug

Expected output

```
Pclass
1    0.629630
2    0.472826
3    0.242363
Name: Survived, dtype: float64
Embarked_S
0    0.506073
1    0.336957
Name: Survived, dtype: float64
FamilySize
0     0.303538
1     0.552795
2     0.578431
3     0.724138
4     0.200000
5     0.136364
6     0.333333
7     0.000000
10    0.000000
Name: Survived, dtype: float64
IsAlone
0    0.505650
1    0.303538
Name: Survived, dtype: float64
Pclass
1    84.154687
2    20.662183
3    13.675550
Name: Fare, dtype: float64
Pclass
1    38.233441
2    29.877630
3    25.140620
Name: Age, dtype: float64
Survived
0    30.626179
1    28.343690
Name: Age, dtype: float64
Survived
0    22.117887
1    48.395408
Name: Fare, dtype: float64
1    136
3    119
2     87
Name: Pclass, dtype: int64
3    372
2     97
1     80
```

```
Name: Pclass, dtype: int64
```

Actual output

```
Pclass
1    0.629630
2    0.472826
3    0.242363
Name: Survived, dtype: float64
Embarked_S
0    0.506073
1    0.336957
Name: Survived, dtype: float64
FamilySize
0     0.303538
1     0.552795
2     0.578431
3     0.724138
4     0.200000
5     0.136364
6     0.333333
7     0.000000
10    0.000000
Name: Survived, dtype: float64
IsAlone
0    0.505650
1    0.303538
Name: Survived, dtype: float64
Pclass
1    84.154687
2    20.662183
3    13.675550
Name: Fare, dtype: float64
Pclass
1    38.233441
2    29.877630
3    25.140620
Name: Age, dtype: float64
Survived
0    30.626179
1    28.343690
Name: Age, dtype: float64
Survived
0    22.117887
1    48.395408
Name: Fare, dtype: float64
1    136
3    119
2     87
Name: Pclass, dtype: int64
3    372
2     97
1     80
```

4.2.8. TITANIC DATASET ANALYSIS AND DATA CLEANING - 4

Youare providedwiththeTitanicdatasetcontaininginformationaboutpassengersontheTitanic.YourtaskistowritePythoncodetoanswerthefollowingquestions based onthedataset.

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

INPUT

```python
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data = pd.get_dummies(data, columns=['Embarked'],
drop_first=True)


# 1. Get the number of survivors by gender
print(data[data['Survived']==1]['Sex'].value_counts())
# 2. Get the number of non-survivors by gender
print(data[data['Survived']==0]['Sex'].value_counts())
# 3. Get the number of survivors by embarked location
print(data[data['Survived']==1]['Embarked_S'].value_counts())
# 4. Get the number of non-survivors by embarked location
print(data[data['Survived']==0]['Embarked_S'].value_counts())
# 5. Calculate the percentage of children (Age < 18) who
survived
print(data[data['Age']<18]['Survived'].mean())
# 6. Calculate the percentage of adults (Age >= 18) who
survived
print(data[data['Age']>=18]['Survived'].mean())
# 7. Get the median age of survivors
print(data[data['Survived']==1]['Age'].median())
# 8. Get the median age of non-survivors
print(data[data['Survived']==0]['Age'].median())
# 9. Get the median fare of survivors
print(data[data['Survived']==1]['Fare'].median())
# 10. Get the median fare of non-survivors
print(data[data['Survived']==0]['Fare'].median())
```

OUTPUT

Average time 0.035 s 35.00 ms | Maximum time 0.035 s 35.00 ms | 1 out of 1 shown test case(s) passed

Test case 1 35 ms Debug

Expected output

```
female    233
male      109
Name: Sex, dtype: int64
male      468
female     81
Name: Sex, dtype: int64
1    217
0    125
Name: Embarked_S, dtype: int64
1    427
0    122
Name: Embarked_S, dtype: int64
0.5398230088495575
0.3810316139767055
28.0
28.0
26.0
10.5
```

Actual output

```
female    233
male      109
Name: Sex, dtype: int64
male      468
female     81
Name: Sex, dtype: int64
1    217
0    125
Name: Embarked_S, dtype: int64
1    427
0    122
Name: Embarked_S, dtype: int64
0.5398230088495575
0.3810316139767055
28.0
28.0
26.0
10.5
```